

SPICKARD, MO, USA

WMO: 725400

Lat: 40.250N

Lon: 93.720W

Elev: 271

StdP: 98.11

Time Zone: -6.00 (NAC)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.1	-24.3	0.4	-16.2	-21.1	0.6	-14.2	12.0	-1.9	11.0	-1.6	3.4	0	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	35.5	23.6	33.6	23.2	31.7	22.9	25.7	31.3	24.8	30.6	23.9	29.8	4.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.8	29.0	23.0	18.4	28.0	22.1	17.4	27.1	80.8	31.1	76.5	30.8	73.0	29.7	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
10.8	9.1	8.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-2.3	0.1	6.0	11.1	17.5	22.4	25.1	23.8	18.6	12.3	4.9	-0.5
	DBStd	10.80	6.20	6.82	6.18	5.18	3.89	3.69	3.11	3.89	4.94	4.87	5.70	6.12
	HDD10.0	1393	382	281	152	47	1	0	0	0	2	31	171	327
	HDD18.3	3074	639	512	383	226	63	8	1	4	57	193	405	584
	CDD10.0	1993	1	2	30	78	232	373	467	428	259	103	17	2
	CDD18.3	631	0	0	2	7	36	131	209	174	64	8	1	0
	CDH23.3	6071	0	0	21	82	282	1227	2164	1664	549	79	3	0
	CDH26.7	2487	0	0	6	19	58	479	971	745	194	15	0	0
Wind	WSAvg	3.9	4.2	3.9	4.7	4.6	4.0	3.6	3.4	3.1	3.5	3.8	4.3	4.0
Precipitation	PrecAvg	906	28	26	62	87	124	108	121	101	102	77	50	37
	PrecMax	1401	100	88	190	176	393	277	542	266	276	211	163	139
	PrecMin	403	0	2	0	16	27	11	19	1	18	4	0	0
	PrecStd	217	23	20	42	44	76	63	100	57	61	47	38	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	18.4	27.3	29.4	30.6	36.1	37.9	38.2	35.0	28.5	23.1	16.9
		MCWB	10.0	10.8	16.1	19.3	21.0	22.8	24.3	24.2	23.6	18.7	15.6	13.1
	2%	DB	10.9	14.2	21.9	25.8	28.2	33.1	35.3	35.5	31.7	25.5	18.9	13.1
		MCWB	6.1	9.1	13.7	17.0	19.8	22.1	23.8	23.6	21.4	17.1	14.1	10.3
	5%	DB	8.1	11.3	18.7	22.8	26.3	31.1	33.5	33.2	28.6	22.9	16.0	9.9
		MCWB	4.7	6.7	12.0	14.5	18.8	22.8	23.4	23.5	21.0	16.1	11.7	6.8
	10%	DB	5.6	8.8	15.8	20.0	24.6	29.3	31.7	30.7	26.4	20.4	13.5	6.8
		MCWB	2.6	5.3	10.6	13.2	17.7	21.8	22.9	22.8	20.0	14.7	9.9	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.6	17.2	19.6	22.6	26.1	27.8	26.2	24.3	20.0	16.8	14.8
		MCDB	13.7	17.5	25.6	27.2	28.7	30.3	32.1	33.3	31.2	26.6	21.2	16.3
	2%	WB	6.8	9.1	14.9	17.6	20.8	24.8	26.3	25.2	23.1	18.3	14.9	10.6
		MCDB	10.2	13.8	20.3	24.2	26.4	29.6	31.4	32.1	29.6	23.3	18.2	12.3
	5%	WB	4.8	7.1	13.0	15.7	19.8	23.8	25.1	24.3	22.1	17.1	12.3	6.6
		MCDB	8.0	11.0	17.9	20.8	25.2	28.9	30.5	30.7	27.2	21.6	15.6	9.7
	10%	WB	3.0	5.4	10.9	13.7	18.5	22.8	24.1	23.5	20.7	15.5	10.0	4.4
		MCDB	5.2	8.5	15.5	18.9	23.1	28.0	30.1	29.3	25.4	19.9	13.2	6.6
Mean Daily Temperature Range		MDBR	9.0	9.4	10.5	11.1	10.6	10.9	11.2	11.1	11.5	11.4	9.6	8.8
	5% DB	MCDBR	12.1	13.3	15.0	15.5	12.7	12.5	12.8	13.8	13.1	14.3	12.6	12.8
		MCWBR	8.8	9.1	9.3	9.2	7.3	6.0	5.5	5.2	6.3	8.3	8.9	9.6
		MCDBR	11.7	12.4	14.1	13.1	10.9	10.9	10.9	11.7	11.0	12.3	10.8	11.8
	MCWBR	8.6	8.8	9.5	8.6	7.1	5.8	5.6	5.3	5.9	7.9	8.4	9.3	
Clear-Sky Solar Irradiance	taub	0.292	0.318	0.338	0.389	0.419	0.421	0.435	0.431	0.381	0.340	0.298	0.289	
	taud	2.441	2.365	2.392	2.266	2.249	2.281	2.240	2.271	2.359	2.470	2.527	2.475	
	Ebn at Noon	876	896	916	884	862	857	841	835	857	862	866	852	
	Edh at Noon	79	99	108	131	136	132	137	129	110	89	72	71	
All-Sky Solar Radiation	RadAvg	2.00	2.90	3.78	4.84	5.68	6.32	6.35	5.54	4.74	3.23	2.23	1.74	
	RadStd	0.20	0.24	0.31	0.33	0.47	0.56	0.45	0.30	0.30	0.36	0.20	0.18	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page